

TOPIC 8/60

Decoder-based Transformers

The architecture behind ChatGPT.

The ultimate text generator.

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THE EVOLUTION

While Google took the "Encoder" half of the Transformer to build BERT (for search), another startup had a different idea.

OpenAI's Bet:

They threw away the Encoder completely. They took only the **Decoder** (the writer half), scaled it up massively, and fed it the entire internet.

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THE SUPERPOWER



Autoregressive Generation

Decoders are built for one singular task: **Predict the next word.**

"Autoregressive" means it feeds its own output back into itself.

It generates word #1. Then it looks at word #1 to generate word #2. Then it looks at words 1 and 2 to generate word #3.

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THE BIG DIFFERENCE

The Blindfold

Encoders read a sentence in both directions at once.
Decoders are strictly unidirectional (left-to-right).

Masked Self-Attention

During training, the Decoder is mathematically blocked ("masked") from seeing future words. It is forced to learn how to predict the unknown future based only on the past.

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THE KING OF DECODERS

GPT

Generative Pre-trained Transformer

GPT models are the most famous examples of Decoder-only architectures.

Other famous Decoder models include Meta's **LLaMA**, Anthropic's **Claude**, and Google's **Gemini** (text generation components).

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THE SCALE

Trillions of Params

Storing World Knowledge

Because Decoders generate answers from scratch, they have to store facts about the world inside their own neural network.

Unlike Encoders (which just analyze the text you give them), Decoders need massive amounts of parameters to remember history, science, coding languages, and logic.

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WHAT ARE THEY GOOD AT?

CONVERSATIONAL AI

Chatbots and Assistants.

Predicting the natural flow of a human conversation and responding dynamically.

CONTENT GENERATION

Writing Code & Essays.

Given a prompt, they effortlessly spin up paragraphs of coherent, creative text.

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THE WEAKNESS

The Cascade Effect

Because Decoders feed their own output back into themselves, a single mistake is devastating.

If it hallucinates one wrong fact in sentence 1, it will treat that lie as absolute truth when generating sentence 2. This leads to confident, cascading hallucinations.

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THE QUICK RECAP

Encoders (BERT)

Reads both ways. Can see the whole sentence. Best at deeply understanding and categorizing text.

Decoders (GPT)

Reads left-to-right. Masked from the future. Best at predicting and creating brand new text.

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Decoder Models

CONCEPT UNLOCKED



The "Writer" of AI



Reads Left-to-Right



The architecture of GPT



Autoregressive Generation

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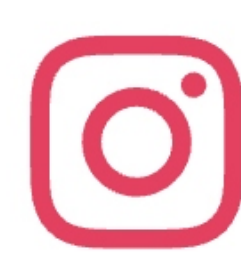
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